PATENT SPECIFICATION

State-Adaptive Compliance Enforcement System
for Decentralised Finance Networks

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1. TECHNICAL FIELD

The present invention relates to systems and methods for enforcing anti-money laundering (AML) and regulatory compliance in decentralised finance (DeFi) transactions using blockchain smart contracts, machine learning risk scoring, and zero-knowledge cryptographic proofs, and more particularly to such systems in which enforcement parameters are computed in real time from a closed-form state-adaptive controller that models the transaction processing network as a multi-body dynamical system with an associated energy landscape.

2. BACKGROUND

2.1. Technical Problems

Decentralised finance protocols process financial transactions on blockchain networks without centralised intermediaries. This architecture creates two interacting technical problems that prior art has not solved in combination.

2.1.1. Pre-Settlement Compliance Enforcement

Existing blockchain analytics platforms (Chainalysis, Elliptic) operate retrospectively: they identify suspicious transactions *after* settlement, by which time irrecoverable value transfer has already occurred. Platform-level KYC/AML checks are bypassed entirely when users transact directly through a DeFi protocol. Supervised fraud detection models require large quantities of labelled fraudulent transactions; in practice fewer than 0.1% of blockchain transactions are labelled as illicit, producing severe class imbalance. Privacy regulations (UK Data Protection Act 2018, EU GDPR) conflict with the traditionally disclosure-intensive compliance model.

2.1.2. Static Enforcement Parameters

Existing compliance systems apply fixed risk thresholds, fixed escrow rules, and fixed proof requirements regardless of the structural state of the underlying transaction network. This is provably suboptimal: above a critical manifold in state space, any constant enforcement configuration necessarily fails to achieve a state-dependent welfare optimum. No prior art derives a closed-form optimal state-adaptive enforcement function as a mapping from a computed energy landscape to numerical thresholds and proof requirements.

2.2. Prior Art Limitations

Prior art known to the applicant includes blockchain analytics retrospective monitoring platforms; exchange-level KYC/AML systems that do not extend to protocol-level transactions; standard supervised fraud detection systems requiring labelled datasets; zero-knowledge privacy protocols (e.g., Zcash) that conceal transactions without implementing compliance; macroprudential risk measures including CoVaR, SRISK, and the Basel III countercyclical capital buffer rule (which applies a constant-friction piecewise-linear function to the credit-to-GDP gap and does not derive a closed-form state-dependent optimal policy); and agent-based financial network simulations lacking closed-form optimal policies and provable stability guarantees. None of the prior art provides the combination of pre-settlement enforcement, weak supervision, closed-form state-dependent threshold and proof-requirement selection, and privacy-preserving verification.

3. SUMMARY OF THE INVENTION

The invention provides a state-adaptive compliance enforcement system and method for DeFi networks whose enforcement configuration is computed in real time by a closed-form controller derived from a dynamical model of the transaction network. The shared inventive concept is the *state-dependent selection of a risk threshold profile and a proof-requirement policy as a function of network instability metrics computed from an energy landscape derived by modelling the network as a multi-body dynamical system.* The primary technical contributions are:

(i) A **four-layer DeFi compliance architecture** comprising interface, compliance orchestration, offline training, and blockchain infrastructure layers that together implement a deterministic compliance lifecycle enforcing policy actions *before* transaction settlement.

(ii) A **weak supervision pipeline** comprising a composite teacher ensemble (autoencoder–gradient boosting, FATF rule-based, graph-structural) generating pseudo-labels from unlabelled blockchain data, and a student model trained on those pseudo-labels, eliminating the need for labelled fraud data.

(iii) A **state-adaptive transaction flow control module** that computes one or more network instability metrics from the transaction graph and aggregate risk dynamics, and selects an enforcement profile from a predetermined set, the selected profile deterministically defining:

   (a) a *risk threshold profile* comprising numerical thresholds for the compliance decision matrix; and

   (b) a *proof-requirement policy* specifying, for at least one risk score tier or transaction class, one or more cryptographic proofs to be verified on-chain as a prerequisite to settlement.

(iv) A **gravitational interaction engine** that models financial institutions as particles in a feature space under a pairwise potential comprising erf-attraction and logarithmic repulsion, with a blind-spot energy coupling, producing a combined systemic energy function $E(X)$.

(v) A **spectral phase transition detector** computing the spectral radius of the interaction matrix and determining whether the network state lies above or below a critical manifold, above which state-adaptive control is provably necessary.

(vi) A **closed-form optimal adaptive policy** $\gamma^*(X)$ derived from the Bellman equation, mapping systemic energy and its gradient to an optimal friction coefficient, which is further mapped to the enforcement profile defining the risk threshold profile and proof-requirement policy.

(vii) A **zero-knowledge proof subsystem** (zkNAF) comprising Groth16 zk-SNARK circuits for sanctions non-membership, risk score range, and KYC verification proofs, enabling privacy-preserving on-chain verification.

(viii) **Smart contract infrastructure** that enforces all policy actions deterministically before settlement, with an immutable on-chain audit trail.

4. DETAILED DESCRIPTION

4.1. System Architecture Overview

The system comprises four hierarchically organised layers:

Layer I — Interface Layer. Client SDKs (TypeScript and Python) and a RESTful API provide the entry points through which external applications submit transaction requests, query risk scores, and verify compliance status. A web dashboard provides visual compliance monitoring.

Layer II — Compliance Orchestration Layer. The core compliance logic resides here. It comprises:

- A *machine learning risk scoring engine* computing a numerical risk score from 167 engineered features (address-level, transaction-level, graph-level) for each submitted transaction request.
- A *graph analytics module* constructing transaction graphs and computing centrality, clustering, PageRank and community features.
- A *sanctions screening module* checking counterparties against OFAC and EU sanctions databases.
- A *state-adaptive transaction flow control module* (the adaptive friction controller, described in §4.5) computing instability metrics and selecting the enforcement profile.
- A *deterministic policy engine* mapping calibrated risk scores to policy actions via a decision matrix with thresholds drawn from the enforced profile.

Layer III — Offline Training Layer. Implements the composite teacher ensemble, pseudo-label generation, student model training, cross-validation, and sigmoid score calibration.

Layer IV — Infrastructure Layer. Blockchain smart contracts, the zkNAF proof subsystem, databases, and distributed file storage for immutable audit records.

4.2. Composite Teacher Weak Supervision Pipeline

The offline training layer constructs three independent teacher models to generate pseudo-labels from unlabelled transaction data:

(a) **Autoencoder–Gradient Boosting Teacher ($s_{\textbf{AE-XGB}}$):** An autoencoder learns a compressed representation of transaction features; a gradient boosting model is trained on reconstruction errors and latent features to estimate anomaly scores.

(b) **FATF Rule-Based Teacher ($s_{\textbf{FATF}}$):** Encodes Financial Action Task Force red-flag indicators (rapid fund movement, structuring patterns, round-number transactions, high-risk jurisdiction involvement) as deterministic scoring rules.

(c) **Graph-Based Teacher ($s_{\textbf{Graph}}$):** Computes suspicion scores from graph-structural properties: proximity to known high-risk addresses, temporal clustering, and unusual connectivity.

The composite teacher score is:

$$s_{\text{teacher}}(x) = w_1 s_{\text{AE-XGB}}(x) + w_2 s_{\text{FATF}}(x) + w_3 s_{\text{Graph}}(x) \tag{1}$$

In the preferred embodiment w_1=0.4, w_2=0.3, w_3=0.3. Binary pseudo-labels are produced by thresholding and used to train an XGBoost student model on 167 engineered features.

4.3. Risk Score Calibration

The raw student model output is recalibrated:

$$\hat{p}_{\text{cal}}(x) = \sigma(k \cdot (s(x) - P_{75})) \tag{2}$$

where σ is the logistic sigmoid, k=30 is a steepness parameter, and P_{75} is the 75th percentile of the training score distribution. The calibrated probability is mapped to an integer score on $[0, 1000]$.

4.4. Compliance Decision Matrix

The deterministic decision matrix maps integer scores to policy actions:

Score Range	Policy Action	Smart Contract Effect
$[0, \tau_1)$	APPROVE	Immediate execution
$[\tau_1, \tau_2)$	APPROVE + MONITOR	Execution; monitoring flag
$[\tau_2, \tau_3)$	REVIEW	Queue for manual review
$[\tau_3, \tau_4)$	ESCROW	Time-locked escrow
$[\tau_4, 1000]$	BLOCK	On-chain rejection

Under the *stable* enforcement profile the thresholds are $(\tau_1, \tau_2, \tau_3, \tau_4) = (200, 400, 600, 800)$. The state-adaptive module selects profiles that adjust these thresholds and impose additional proof requirements as a function of computed network instability.

4.5. State-Adaptive Transaction Flow Control Module

4.5.1. Instability Metrics

The module computes one or more network instability metrics from:

(a) the spectral radius or centrality concentration of the transaction graph constructed from recent on-chain activity over a sliding time window;

(b) a temporal burst indicator computed from the transaction count process; and/or

(c) a shift of the calibrated risk score distribution relative to a reference distribution.

4.5.2. Enforcement Profile Selection

Based on the computed metrics the module selects one profile from a discrete set (e.g., *stable*, *stressed*, *critical*). Each profile deterministically defines:

(a) a *risk threshold profile* $(\tau_1, \tau_2, \tau_3, \tau_4)$ for the decision matrix; and

(b) a *proof-requirement policy* specifying, for at least one risk score tier or transaction class, which of the zkNAF proofs (sanctions non-membership, risk range, KYC) must be verified on-chain prior to settlement.

For a fixed system state the selected profile and all resulting decisions are deterministic and reproducible.

4.5.3. Macro-Level Systemic Energy Computation

When financial metric data for multiple institutions is available (e.g., aggregated on-chain liquidity metrics, cross-protocol risk exposures), the module further computes network instability using a gravitational interaction model. The system models N institutions as particles in $\mathbb{R}^d$ governed by:

$$\dot{X} = -\alpha(X - \mu\mathbf{1}^\top) - \nabla_X \Phi_{\text{pair}}(X) - \sum_k \beta_k D\Phi_k(X)^\top D_k(X) \tag{3}$$

where $X \in \mathbb{R}^{N\times d}$ is the state matrix, α is a mean-reversion rate, Φ_{pair} is the pairwise interaction potential, and Φ_k are external field operators.

4.5.4. Pairwise Interaction Potential

The pairwise potential between institutions i and j at distance $r = \|x_i - x_j\|$ is:

$$\phi(r) = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2} \operatorname{erf}\left(\frac{r}{\sigma}\right) - \gamma\lambda_{\text{rep}} \cdot \log(r + \varepsilon) \tag{4}$$

The erf-attraction is bounded (unlike gravitational $1/r$ attraction), and the logarithmic repulsion prevents institution collapse. In the preferred embodiment: α=0.10, γ=1.00, σ=1.00, λ_{rep}=0.10, ε=10^{-5}.

4.5.5. Blind-Spot Energy Module

A complementary energy function is computed from four deviation channels relative to reference distributions fitted on a normal period:

(a) enstrophy anomaly δ_C — rotational energy deviation;

(b) spectral gap δ_G — structural fragility indicator;

(c) alignment anomaly δ_A — co-movement deviation; and

(d) temporal novelty δ_T — drift from reference state patterns.

The blind-spot energy is:

$$E_{\mathrm{BS}}(X) = \|(X - \mu_0)^\top \Sigma_0^{-1}(X - \mu_0)\|_F \tag{5}$$

where μ_0 and Σ_0 are the reference mean and covariance estimated by Ledoit–Wolf Oracle shrinkage.

4.5.6. Critical Manifold

A critical manifold $\mathcal{C}$ separates regimes in which constant enforcement parameters are adequate from regimes in which they are provably suboptimal:

Theorem (Spectral Phase Transition):

$$\mathcal{C} = \{X : \lambda_{\max}(\rho(X) \cdot \ell(X) \cdot W) = 1\} \tag{6}$$

where $\rho(X)$ is the spectral radius of the inter-institution correlation matrix, $\ell(X)$ is the system leverage factor, and W is the weighted adjacency matrix. Above $\mathcal{C}$ state-adaptive friction is both necessary and sufficient for optimality.

4.5.7. Closed-Form Optimal Adaptive Policy

Theorem (Optimal Friction): The optimal state-dependent friction coefficient solving the associated Bellman equation is:

$$\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa + \beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X) + \theta} \tag{7}$$

where $E(X)$ is the combined systemic energy, κ is the economic cost parameter, β is the systemic damage coefficient, and $\theta = \kappa/\beta$. The policy satisfies $\gamma^* \to 0$ as $E \to 0$ (no friction when stable) and $\gamma^* \to 1$ as $E \to \infty$ (maximum friction under severe instability).

The scalar $\gamma^*(X) \in [0, 1]$ is used to select the enforcement profile: below a lower threshold γ_1 the *stable* profile is active; between γ_1 and γ_2 the *stressed* profile is active; above γ_2 the *critical* profile is active.

4.5.8. Navier–Stokes Analogy (Enhanced Embodiment)

Institution position changes $\dot{X}$ are treated as a velocity field. The velocity gradient tensor is decomposed into strain S and vorticity Ω. Enstrophy $\mathcal{E} = \frac{1}{2}\|\omega\|_F^2$ quantifies rotational energy. The Beale–Kato–Majda criterion adapted to financial networks serves as a singularity indicator. The adaptive friction coefficient is reinterpreted as an adaptive viscosity: $\nu = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}))$. Minsky regime classification (hedge, speculative, Ponzi) emerges from the cosine alignment between the gravitational gradient and the blind-spot gradient.

4.6. Zero-Knowledge Proof Subsystem (zkNAF)

The zkNAF subsystem implements three Groth16 zk-SNARK proof circuits:

Sanctions Non-Membership Proof. The prover demonstrates that a blockchain address is not a member of a Merkle tree of sanctioned addresses (using the Poseidon hash function) without revealing the address. The verifier checks the proof against the published Merkle root.

Risk Range Proof. The prover demonstrates: $s_{\min} \leq s(x) \leq s_{\max}$ without revealing the exact score $s(x)$.

KYC Verification Proof. The prover demonstrates completion of Know Your Customer verification with an accredited provider without revealing identity data.

Each proof has a 24-hour time-to-live. Verification is performed on-chain by a dedicated ZK verifier smart contract that performs the bilinear pairing check and records the result with a timestamp.

4.7. Smart Contract Architecture

- **Core Contract:** Implements the deterministic decision matrix on-chain; registers compliance decisions as an immutable audit record.
- **Escrow Contract:** Holds funds in time-locked escrow for ESCROW-policy transactions; supports release and dispute escalation.
- **ZK Verifier Contract:** Verifies Groth16 proofs; stores binary verification results and timestamps.
- **Cross-Chain Bridge Contract:** Requires compliance verification on both source and destination chains before releasing bridged funds.
- **Dispute Resolution Contract:** Structured escalation and release process for escrowed transactions under dispute.

All enforcement decisions execute *before* the corresponding transaction is settled on the blockchain network.

4.8. Feature Engineering

The risk scoring engine operates on 167 features across three categories:

- **Address-level:** Transaction count, total value, mean and variance of amounts, address age, unique counterparty count, temporal activity patterns.
- **Transaction-level:** Value, gas parameters, input data characteristics, token transfer indicators.
- **Graph-level:** Degree centrality, betweenness centrality, clustering coefficient, PageRank, community membership, local graph density.

4.9. Performance Characteristics

DeFi Compliance System (trained on 2.64 million Ethereum transactions):

Metric	Value
Meta-Ensemble ROC-AUC (test)	1.0000
Meta-Ensemble ROC-AUC (5-fold CV)	0.9965 ± 0.0049
Meta-Ensemble PR-AUC	0.9999
Transaction-Level XGBoost AUC	0.9878 F1: 0.901
Cross-domain mean AUC (9 datasets)	0.9962
Escrow contract gas cost	287,890 gas

Adaptive Friction Controller (FDIC SDI data, 140 quarters):

Configuration	N	AUROC	GFC Lead
Institution-type (7 classes)	7	≥ 0.70	6 quarters
Bank-level (individual)	30	0.805	6 quarters
G-SIB cross-border	20	0.781	6 quarters

Super-criticality prevalence: 68.6% of sample above $\mathcal{C}$. GFC welfare loss of inaction: 19.57 percentage points consumption-equivalent. Adversarial robustness: 5 of 12 attack types never break detection across all configurations.

5. CLAIMS

The following claims define the scope of protection sought. Where a claim refers to another claim, the dependency is by claim number. Features of any dependent claim may be combined with any independent claim or with other dependent claims.

1. A computer-implemented transaction compliance system for decentralised finance networks, the system comprising:

(a) an interface layer configured to receive transaction requests from client applications;

(b) a compliance orchestration layer comprising:

(i) a machine learning risk scoring engine configured to compute a numerical risk score for each transaction request from engineered features comprising address-level, transaction-level, and graph-level attributes;

(ii) a graph analytics module configured to construct and analyse transaction graphs and compute graph-level features;

(iii) a sanctions screening module configured to check transaction counterparties against maintained sanctions databases;

(iv) a state-adaptive transaction flow control module configured to compute one or more network instability metrics from recent transaction activity and to select, from a predetermined set of enforcement profiles, an enforcement configuration comprising:

(a) a risk threshold profile defining numerical thresholds for a deterministic compliance decision matrix; and

(b) a proof-requirement policy defining, for at least one risk score tier or transaction class, one or more cryptographic proofs required to be verified on-chain as a prerequisite to settlement;

and

(v) a deterministic policy engine configured to map computed risk scores to policy actions according to the selected risk threshold profile, and to enforce the proof-requirement policy by rejecting transactions for which the required proofs are missing or fail on-chain verification;

(c) an offline training layer comprising a composite teacher ensemble that generates pseudo-labels from unlabelled blockchain transaction data using a weighted combination of at least two independent teacher models, and a student model trained on those pseudo-labels; and

(d) an infrastructure layer comprising smart contracts deployed on a blockchain network configured to enforce the determined policy actions before settlement, including a core contract, an escrow contract, and a zero-knowledge proof verifier contract;

wherein all compliance decisions are enforced before the corresponding transaction is settled on the blockchain network.

2. A computer-implemented method of enforcing compliance in decentralised finance transactions, the method comprising:

(a) receiving a transaction request comprising at least a sender address, a recipient address, and a transaction value;

(b) computing a risk score by applying a trained machine learning model to a feature vector comprising address-level, transaction-level, and graph-level features;

(c) calibrating the risk score using a sigmoid recalibration function anchored at a percentile of a training score distribution;

(d) computing one or more network instability metrics from recent transaction activity and selecting an enforcement profile from a predetermined set, the selected profile deterministically defining:

(i) a risk threshold profile for a deterministic decision matrix; and

(ii) a proof-requirement policy specifying one or more cryptographic proofs to be verified on-chain as a prerequisite to settlement for at least one risk score tier or transaction class;

(e) mapping the calibrated risk score to a policy action using the decision matrix with thresholds from the selected risk threshold profile;

(f) enforcing the proof-requirement policy by verifying the specified proofs on-chain prior to settlement, and applying a non-approval action when required proofs are absent or fail verification;

(g) executing a corresponding smart contract function on a blockchain network before settlement; and

(h) recording the compliance decision and policy action on-chain as an immutable audit record.

3. A computer-implemented system for monitoring systemic risk in a financial network and outputting an enforcement configuration for a transaction processing network, the system comprising:

(a) a gravitational interaction engine configured to model a plurality of financial institutions as particles in a multi-dimensional feature space interacting through a pairwise potential comprising a bounded erf-attraction term and a logarithmic repulsion term;

(b) a spectral monitoring module configured to compute the spectral radius of an interaction matrix and to determine whether the network state falls above or below a critical manifold defined by the condition that the spectral radius equals one;

(c) a blind-spot energy module configured to compute, from reference distributions fitted on a designated normal period, a complementary energy function comprising at least an enstrophy anomaly channel, a spectral gap channel, an alignment anomaly channel, and a temporal novelty channel; and

(d) an adaptive policy module configured to compute a state-dependent friction coefficient from the combined systemic energy and its gradient and to output an enforcement configuration comprising:

(i) a risk threshold profile defining numerical thresholds for transaction approval decisions in a transaction processing network; and

(ii) a proof-requirement policy specifying, for at least one threshold tier or transaction class, one or more cryptographic proofs required to be verified as a prerequisite to settlement.

4. A system according to claim 1 or claim 3, wherein the pairwise potential is defined as:

$$\phi(r) = \gamma \cdot \frac{\sigma\sqrt{\pi}}{2}\,\mathrm{erf}(r/\sigma) - \gamma\lambda_{\mathrm{rep}} \cdot \log(r + \varepsilon)$$

where r is the inter-institution distance, γ is an interaction strength, σ is an interaction scale, λ_{rep} is a repulsion strength, and ε is a regularisation constant.

5. A method according to claim 2 or a system according to claim 3, wherein the state-dependent friction coefficient is:

$$\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa + \beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X) + \theta}$$

where $E(X)$ is the combined systemic energy, $\|\nabla E\|$ is the gradient norm, κ is an economic cost parameter, β is a systemic damage coefficient, and $\theta = \kappa/\beta$.

6. A system or method according to claims 1 to 5, wherein the enforcement profile is selected from a discrete set of profiles, and wherein the selection is determined by whether the computed friction coefficient $\gamma^*(X)$ lies between predetermined lower and upper thresholds.

7. A system or method according to claims 1 to 6, wherein the proof-requirement policy comprises, for at least one enforcement profile, requiring at least one of: a sanctions non-membership proof, a risk range proof, or a KYC verification proof.

8. A system or method according to claims 1 to 7, wherein the zero-knowledge proofs are Groth16 zk-SNARK proofs, the sanctions non-membership proof uses a Merkle tree constructed with the Poseidon hash function, and each proof has a time-to-live after which it must be regenerated.

9. A system or method according to claims 1 to 8, wherein the composite teacher ensemble comprises an autoencoder–gradient boosting teacher, a rule-based FATF teacher, and a graph-structural teacher, and the composite score is a weighted sum of their outputs.

10. A method according to claim 2, wherein the sigmoid recalibration function is:

$$\hat{p}_{\mathrm{cal}}(x) = \sigma(k \cdot (s(x) - P_{75}))$$

where σ is the logistic sigmoid, k is a steepness parameter, and P_{75} is the 75th percentile of the training score distribution.

11. A system or method according to claims 1 to 10, wherein selecting the enforcement profile further deterministically defines at least one of: a transaction rate limit, a queue delay, an escrow duration, an escrow release condition window, or a maximum transaction value limit.

12. A system according to claim 3, further comprising a Navier–Stokes analogy module configured to compute a velocity gradient tensor from institution position changes, decompose it into strain rate and vorticity components, compute enstrophy as $\mathcal{E} = \frac{1}{2}\|\omega\|_F^2$, and evaluate a Beale–Kato–Majda blow-up criterion.

13. A system according to claim 12, wherein the adaptive friction coefficient is applied as an adaptive viscosity $\nu = \nu_0(1 + \gamma^*(E_{\mathrm{BS}}))$ in the Navier–Stokes analogy.

14. A system according to claim 3, further comprising a Minsky regime classification module configured to classify the network into hedge, speculative, and Ponzi regimes based on the cosine alignment between the gravitational gradient and the blind-spot energy gradient.

15. A system or method according to claims 3 to 14, wherein the spectral radius is computed using power iteration with finite-difference Hessian-vector products at $O(N^2 d \log N)$ complexity per time step.

16. A system according to claim 1, further comprising a cross-chain bridge contract requiring compliance verification on both a source chain and a destination chain before releasing bridged funds.

17. A system or method according to claims 3 to 15, wherein a welfare cost of not implementing adaptive friction is computed as:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\mathrm{inaction}}/(\sigma\theta T)}\right)$$

and the system operates as an online algorithm with $O(\sqrt{T})$ cumulative regret relative to the best fixed policy in hindsight.

6. DRAWINGS

[To be filed separately: (1) System architecture diagram showing the four compliance layers; (2) Flow diagram of the compliance decision lifecycle with proof-gating; (3) Schematic of the zkNAF circuits and on-chain verification flow; (4) Block diagram of the composite teacher weak supervision pipeline; (5) Diagram of the GravityEngine N-body interaction model; (6) Plot of the erf-log pairwise potential function; (7) Diagram of the critical manifold $\mathcal{C}$ in state space with enforcement profile regions; (8) Time series of the GFC early warning signal at 2007-Q1; (9) Navier–Stokes analogy schematic; (10) Minsky regime classification diagram.]

ABSTRACT

A state-adaptive compliance enforcement system for decentralised finance networks enforces regulatory policy before transaction settlement. A state-adaptive transaction flow control module computes network instability metrics and selects an enforcement profile; the selected profile deterministically defines a risk threshold profile for the decision matrix and a proof-requirement policy gating settlement. A closed-form optimal friction coefficient derived from a multi-body gravitational model of the financial network drives profile selection. A machine learning risk scoring engine trained via composite teacher weak supervision scores transactions; blockchain smart contracts enforce outcomes before settlement. A zero-knowledge proof subsystem enables privacy-preserving on-chain verification of sanctions status, risk range and KYC without revealing underlying data. The system detects the Global Financial Crisis six quarters in advance on FDIC data and achieves ROC-AUC of 0.9965 across 2.64 million Ethereum transactions.